Multi-Area OSPF Setup Guide

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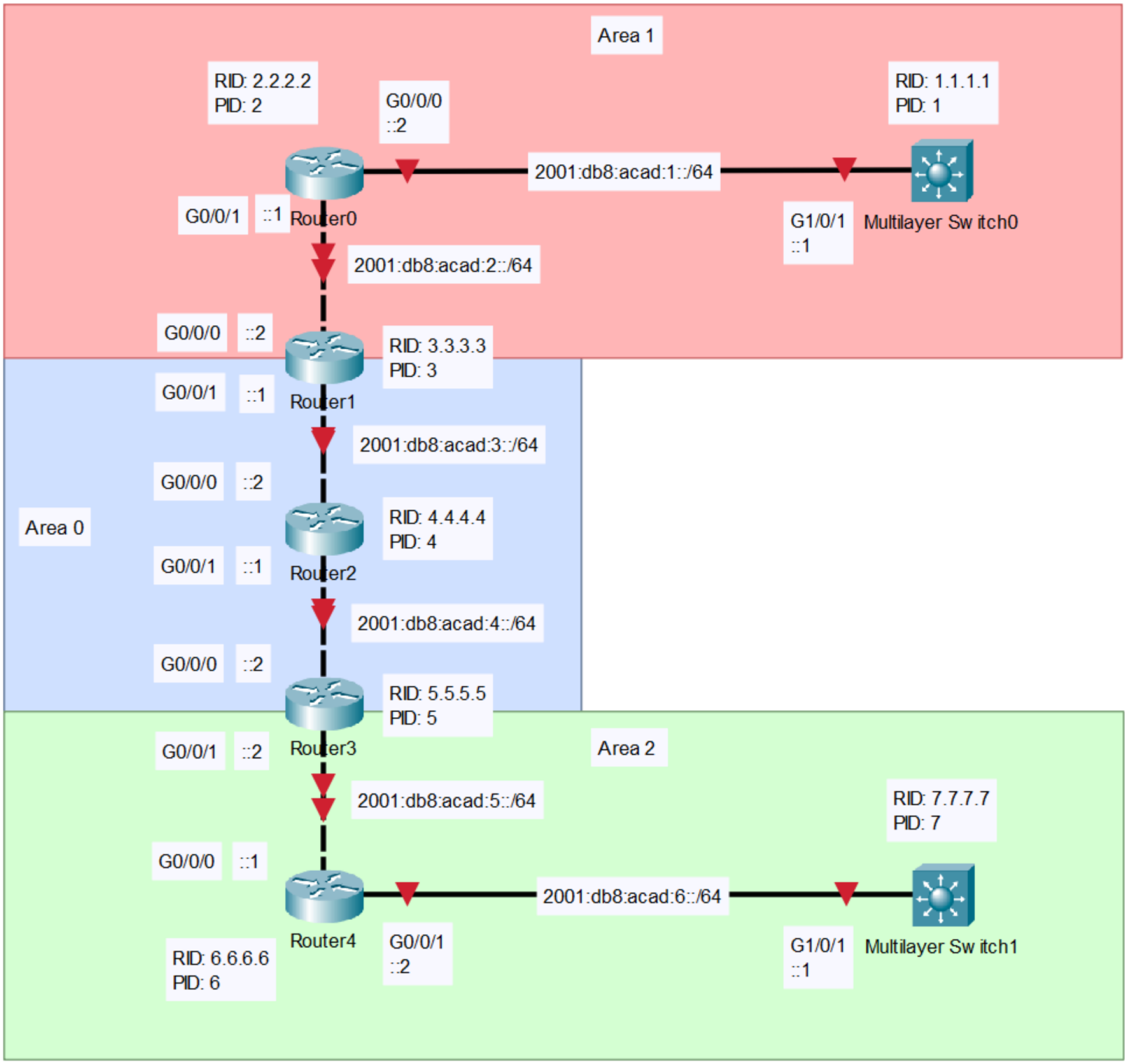
Background

OSPF, a routing protocol designed originally in 1989, produced various offshoots over the years in terms of various use cases and versions, especially a 1999 version targeting IPv6 routing. OSPF, standing for Open Shortest Pathway First, still an important routing protocol today, works by assigning costs to routing pathways called metrics, and sending this information through the network. The lowest cost pathway to a destination is kept, while the others are discarded, although due to its nature as a dynamic routing protocol, it can adjust to changes in the network if configured correctly. Originally OSPF was created to bypass the limitations of distance-vector based routing protocols like RIP (Routing Information Protocols.) The main difference between the two, is that distance vector based protocols configure their routing, by calculating the distance to any other router based solely on their neighbors, but usually has the issue of persistent routing loops. Link-state protocols like OSPF instead determine link cost by figuring out the whole network's configuration, and calculating internally the best routes, before discarding the inefficient routes, and due to the shared information, can adjust the route on the fly to reconfigure due to changes in topology, but suffers from more network traffic due to the frequent link-state updates.

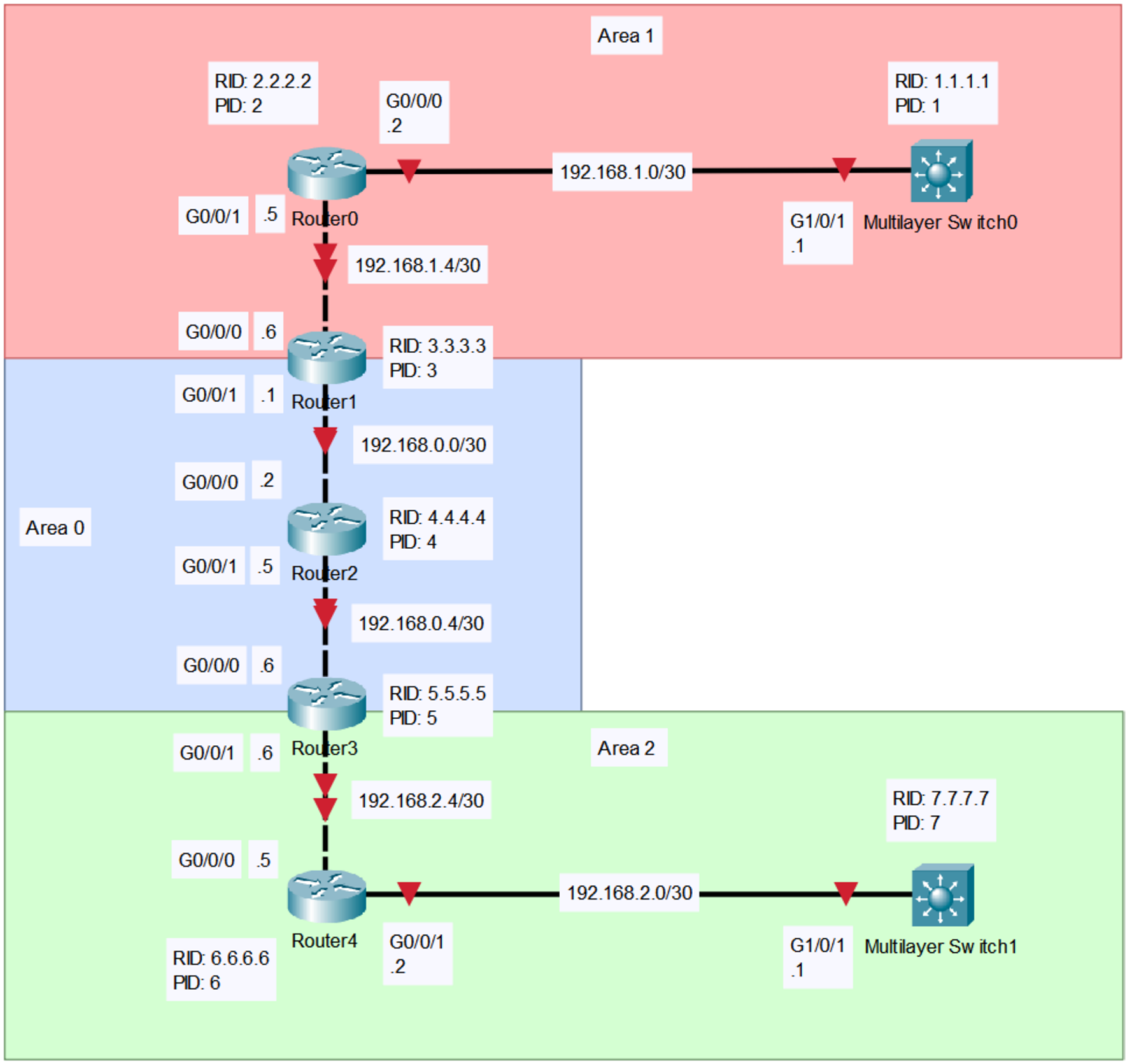
In order to mollify the issues involved with these frequent link information updates, Multi-Area OSPF was designed. This sub-protocol reduces the overhead, by separating the network into multiple smaller areas which helps reduce the amount of updates sent between the various devices. This function, although not too useful when dealing with small networks, is key when dealing with larger ones due to the increased bandwidth available between networks due to the lower number of messages sent due to updates. It functions through the use of a backbone, or a central hub through which all other inter-area traffic must travel through, and the subareas, all sending out brief packets of information to update their status. Each subarea acts the same as the whole network would in standard use OSPF, in that they all constantly advertise to each other, and each area stores and maintains a Link State Database (LSDB.) This is especially important because, if multiple devices or ports go down at the same time, the following update storm is kept to a couple devices in a limited area, preventing the entire network from being delayed or slowed down, as would happen otherwise.

Topologies

IPv6 Topology



IPv4 Topology



Configurations

MLS0

```
version 16.9
hostname MLS0
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
This device has been configured for Mutli-Area OSPF $
ip routing
ipv6 unicast-routing
interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.1.1 255.255.255.252
 ip ospf 1 area 1
 no shutdown
 ipv6 address 2001:db8:acad:1::1/64
 ipv6 ospf 1 area 1
router ospfv3 1
 router-id 1.1.1.1
router ospf 1
 router-id 1.1.1.1
```

MLS1

```
version 16.9
hostname MLS1
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ip routing
ipv6 unicast-routing
interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.2.1 255.255.255.252
 ip ospf 7 area 2
 no shutdown
 ipv6 address 2001:db8:acad:6::1/64
 ipv6 ospf 7 area 2
router ospfv3 7
 router-id 7.7.7.7
router ospf 7
 router-id 7.7.7.7
```

Router 0

```
version 16.9
hostname Router0
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 2
 router-id 2.2.2.2
router ospf 2
 router-id 2.2.2.2
interface GigabitEthernet0/0/0
 ip address 192.168.1.2 255.255.255.252
 ip ospf 2 area 1
 no shutdown
 ipv6 address 2001:db8:acad:1::2/64
 ipv6 ospf 2 area 1
interface GigabitEthernet0/0/1
 ip address 192.168.1.5 255.255.255.252
 ip ospf 2 area 1
 no shutdown
 ipv6 address 2001:db8:acad:2::1/64
 ipv6 ospf 2 area 1
```

Router 1

```
version 16.9
hostname Router1
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 3
 router-id 3.3.3.3
router ospf 3
 router-id 3.3.3.3
interface GigabitEthernet0/0/0
 ip address 192.168.1.6 255.255.255.252
 ip ospf 3 area 1
 no shutdown
 ipv6 address 2001:db8:acad:2::2/64
 ipv6 ospf 3 area 1
interface GigabitEthernet0/0/1
 ip address 192.168.0.1 255.255.255.252
 ip ospf 3 area 0
 no shutdown
 ipv6 address 2001:db8:acad:3::1/64
 ipv6 ospf 3 area 0
```

Router 2

```
version 16.9
hostname Router2
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 4
 router-id 4.4.4.4
router ospf 4
 router-id 4.4.4.4
interface GigabitEthernet0/0/0
 ip address 192.168.0.2 255.255.255.252
 ip ospf 4 area 0
 no shutdown
 ipv6 address 2001:db8:acad:3::2/64
 ipv6 ospf 4 area 0
interface GigabitEthernet0/0/1
 ip address 192.168.0.5 255.255.255.252
 ip ospf 4 area 0
 no shutdown
 ipv6 address 2001:db8:acad:4::1/64
 ipv6 ospf 4 area 0
```

Router 3

```
version 16.9
hostname Router3
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 5
 router-id 5.5.5.5
router ospf 5
 router-id 5.5.5.5
interface GigabitEthernet0/0/0
 ip address 192.168.0.6 255.255.255.252
 ip ospf 5 area 0
 no shutdown
 ipv6 address 2001:db8:acad:4::2/64
 ipv6 ospf 5 area 0
interface GigabitEthernet0/0/1
 ip address 192.168.2.6 255.255.255.252
 ip ospf 5 area 2
 no shutdown
 ipv6 address 2001:db8:acad:5::2/64
 ipv6 ospf 5 area 2
```

Router 4

```
version 16.9
hostname Router4
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 6
 router-id 6.6.6.6
router ospf 6
 router-id 6.6.6.6
interface GigabitEthernet0/0/0
 ip address 192.168.2.5 255.255.255.252
 ip ospf 6 area 2
 no shutdown
 ipv6 address 2001:db8:acad:5::1/64
 ipv6 ospf 6 area 2
interface GigabitEthernet0/0/1
 ip address 192.168.2.2 255.255.255.252
 ip ospf 6 area 2
 no shutdown
 ipv6 address 2001:db8:acad:6::2/64
 ipv6 ospf 6 area 2
```

Proof

MLS1#show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set
```

```
      192.168.0.0/30 is subnetted, 2 subnets
O IA     192.168.0.0 [110/4] via 192.168.2.2, 00:16:01,
GigabitEthernet1/0/1
O IA     192.168.0.4 [110/3] via 192.168.2.2, 00:16:01,
GigabitEthernet1/0/1
      192.168.1.0/30 is subnetted, 2 subnets
O IA     192.168.1.0 [110/6] via 192.168.2.2, 00:16:01,
GigabitEthernet1/0/1
O IA     192.168.1.4 [110/5] via 192.168.2.2, 00:16:01,
GigabitEthernet1/0/1
      192.168.2.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.2.0/30 is directly connected,
GigabitEthernet1/0/1
L        192.168.2.1/32 is directly connected,
GigabitEthernet1/0/1
O        192.168.2.4/30
```

```
MLS1#show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       R - RIP, D - EIGRP, EX - EIGRP external, ND - ND Default
       NDp - ND Prefix, DCE - Destination, NDr - Redirect, O -
OSPF Intra
       OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1
- OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2
OI  2001:DB8:ACAD:1::/64 [110/6]
     via FE80::B6A8:B9FF:FE01:BC61, GigabitEthernet1/0/1
OI  2001:DB8:ACAD:2::/64 [110/5]
     via FE80::B6A8:B9FF:FE01:BC61, GigabitEthernet1/0/1
OI  2001:DB8:ACAD:3::/64 [110/4]
     via FE80::B6A8:B9FF:FE01:BC61, GigabitEthernet1/0/1
OI  2001:DB8:ACAD:4::/64 [110/3]
     via FE80::B6A8:B9FF:FE01:BC61, GigabitEthernet1/0/1
O   2001:DB8:ACAD:5::/64 [110/2]
     via FE80::B6A8:B9FF:FE01:BC61, GigabitEthernet1/0/1
C   2001:DB8:ACAD:6::/64 [0/0]
     via GigabitEthernet1/0/1, directly connected
L   2001:DB8:ACAD:6::1/128 [0/0]
     via GigabitEthernet1/0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

```
MLS1#show ip ospf border-routers

            OSPF Router with ID (7.7.7.7) (Process ID 7)


                Base Topology (MTID 0)

Internal Router Routing Table
Codes: i - Intra-area route, I - Inter-area route

i 5.5.5.5 [2] via 192.168.2.2, GigabitEthernet1/0/1, ABR, Area
2, SPF 3



MLS1#show ipv6 ospf border-routers

            OSPFv3 Router with ID (7.7.7.7) (Process ID 7)

Codes: i - Intra-area route, I - Inter-area route

i 5.5.5.5 [2] via FE80::B6A8:B9FF:FE01:BC61,
GigabitEthernet1/0/1, ABR, Area 2, SPF 6



MLS1#show ip ospf database router

            OSPF Router with ID (7.7.7.7) (Process ID 7)

                Router Link States (Area 2)

  LS age: 1073
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 5.5.5.5
  Advertising Router: 5.5.5.5
  LS Seq Number: 8000000D
  Checksum: 0xD04A
  Length: 36
  Area Border Router
  Number of Links: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.2.6
```

```
      (Link Data) Router Interface address: 192.168.2.6
       Number of MTID metrics: 0
        TOS 0 Metrics: 1
```

MLS1#show ipv6 ospf database router

```
            OSPFv3 Router with ID (7.7.7.7) (Process ID 7)

                Router Link States (Area 2)

  Routing Bit Set on this LSA
  LS age: 1089
  Options: (V6-Bit, E-Bit, R-Bit, DC-Bit)
  LS Type: Router Links
  Link State ID: 0
  Advertising Router: 5.5.5.5
  LS Seq Number: 8000000B
  Checksum: 0xFBC1
  Length: 40
  Area Border Router
  Number of Links: 1

    Link connected to: a Transit Network
      Link Metric: 1
      Local Interface ID: 7
      Neighbor (DR) Interface ID: 7
      Neighbor (DR) Router ID: 5.5.5.5
```

Problems

One of the earliest issues we had was that we couldn't console into the devices, or at least the supposed console ports didn't work as they should. It turned out that there were two problems involved there, one was that the console cables into our own PC's were disconnected, and the second was that the hub wasn't working. In the end we reconnected the console cables to our PC's after much searching, and just decided to ignore the hub and console directly into the routers and switches and that allowed us to get onto configuration. Another major issue was that the first ping test from Router 0 to Router 4 failed, until we checked IPv6 addressing, and found that there were duplicate IPv6 addresses on links on Router 1 and Router 2. After that, we also deleted some of the copied configuration that would just throw up error messages and didn't actually help with any of the setup. The third problem we found was that the MLS Switch wouldn't allow routing or IP addressing on the interfaces, which actually turned out to be due to the fact that on layer 3 switches, you need to manually enable IP routing using `ip routing`. The final problem that we had to deal with is that IPv6 stopped working all of a sudden, especially that when we were pasting in the configurations, the router would return the error message: OSPF Process <PID> failed to allocate unique router-id and cannot start. After trying a few things, like making the OSPF process ID's on the same router different, but even that didn't work. Fortunately, after rebooting and reconfiguring the routers, the problem disappeared, so we have no idea what happened or why it happened.

Conclusion

This lab was rather simple, seeing as we already had a strong base on this setup process, as well as the many online resources available to aspiring network engineers. The basic steps were really setting up the IP addresses, opening the ports, and setting up all the other basic configurations, like banner, and password keys if that tickles your fancy. Then you needed to setup the OSPF, and OSPFv3 router ID's, and then assign them to different ports under their respective areas, and then if you've done all that right, you should be able to ping from one end of the configuration to the other. In fact however, the research for this writeup was the most enlightening part of this lab, the doing of which, allowed me to learn what is so special about link-state routing protocols like OSPF, as opposed to other kinds like distance-vector. The whole thing as mostly painless, although there was a rather large amount of vexation around the inability of our PC's to console into the routers, mostly due to the exasperation involved in trying to follow an individual cable through a mess of ten to twenty other cables in order to find which cable needs to be reconnected, but otherwise it was all clear.